Maha Touri

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Profile

Double-degree engineering student in Mechatronics and Robotic Systems seeking a position as a Robotics, Numerical Simulation, or Embedded Systems Engineer.

Education

SeaTech Engineering School | Double Degree in Mechatronics and Robotics — 2024 – 2026
INSA Hauts-de-France | Engineering Degree in Mechatronic Systems — 2022 – 2026

Experience

Robotics Control and Navigation Engineering Intern | Nimbl'Bot, Bordeaux — March 2026 – Present

- Development of a ROS2 control framework for a mobile manipulator combining the Spot quadruped (Boston Dynamics) with a 25-DOF hyper-redundant robotic arm, using Placo and Pinocchio for kinematics. Leveraged Spot's onboard cameras for object detection with YOLOv5 and explored Reinforcement Learning approaches for robotic control in inspection scenarios.

Medical Robotics Engineering Intern | CRTA, Croatia — April 2025 – August 2025

- Programming and control of KUKA robotic arms for ultrasound and biopsy procedures. Optimized robot base placement using genetic algorithms to reduce singularities and joint limits, while integrating the ROS2 package `reach_ros2` to improve manipulability.
- Simulation of patient motion using a KUKA robot and development of a real-time tracking strategy controlled by a PI controller through a second robot. Designed and 3D-printed a prostate phantom holder using SolidWorks.

Robotics R&D Intern | IME (Military Institute of Engineering), Brazil — September 2023 – January 2024

- Developed a digital twin of a robotic production line with a Yaskawa robotic arm using ROS (Gazebo, MoveIt, RViz) to optimize motion planning and detect collisions before deployment.

Projects and Certifications

- **RoboCup Middle Size League**: Participated in the international RoboCup competition with Robot Club Toulon. Contributed to the development of the YOLOv8-based visual perception system. Achieved the title of RoboCup 2026 MSL World Runners-up.
- **Scientific Publication and RAAD Conference**: Co-authored the scientific paper "*Optimization of Robot Base Placement for Prostate Biopsy Planning*" and presented the work at the international RAAD conference.
- **Autonomous Robot Project**: Programmed a mobile robot using a dsPIC33 microcontroller in C (XC16): PWM motor control, IR range sensors, UART serial communication, timers and interrupts, and implementation of a finite-state machine for autonomous navigation.
- **Autonomous Navigation with TurtleBot2**: Developed an autonomous navigation system using TurtleBot2 and ROS, implementing a finite-state machine for obstacle avoidance based on real-time LiDAR and bumper sensor data.
- **SWARMz4 Project**: Participated in the SWARMz4 competition by developing autonomous swarm strategies for drones using ROS2/Gazebo and a Qt interface.

Technical Skills

- Programming: C, C++, Python (OpenCV, scikit-learn, Placo, Pinocchio)
- Robotics Tools: ROS2 (MoveIt/Gazebo/RViz), MATLAB/Simulink, CATIA, SolidWorks, Fusion 360, Eagle, Git, Ubuntu

Soft Skills

- Open-mindedness, analytical thinking, autonomy, and proactivity
- Communication and teamwork in multicultural environments
- Leadership and responsibility management (Treasurer of Robot Club Toulon)

Languages

- English: TOEIC 940/990
- Spanish: Intermediate

Interests

- Basketball, guitar, pottery